

The Number of Tilings of a Block with Blocks

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We consider a problem posed to us by our colleague, George Nagy. A *block* is a rectangle with positive integer sides. How many tilings of an $k \times n$ block can be obtained by a divide-and-conquer method of tiling? In the divide-and-conquer method a tiling is obtained by repeating the following process a finite number of times: Select a block already in the tiling and subdivide it into two smaller blocks. Let $g(k, n)$ and $d(k, n)$ be the number of tilings of a $k \times n$ block of the general and the divide-and-conquer types, respectively. Our main result is that for a fixed k both of the sequences

$$g_k = (g(k, n): n = 1, 2, \dots), \quad d_k = (d(k, n): n = 1, 2, \dots)$$

are rational; that is, they satisfy linear homogeneous difference equations with constant coefficients. We show how to calculate the coefficients for these difference equations, and use standard methods to obtain asymptotic formulas for $g(k, n)$ and $d(k, n)$ for fixed k as n tends to infinity. The limits

$$\lim_{k, n \rightarrow \infty} (g(k, n))^{1/kn} = \gamma, \quad \lim_{k, n \rightarrow \infty} (d(k, n))^{1/kn} = \delta$$

exist. Numerical evidence suggests that γ and δ are different, and if this is true, $d(k, n)/g(k, n)$ tends to 0 exponentially as k and n tend to infinity.

1. INTRODUCTION

Our colleagues George Nagy and Sharad Seth have recently studied the problem of how to organize information gained by optically scanning a printed page [7]. Such a page is a rectangle subdivided into smaller rectangles. We call a $k \times n$ rectangle having positive integer sides k, n a *block*. The page and its subrectangles of print is modelled by a block cut into sub-blocks; a block subdivided into sub-blocks is said to be *tilled* with blocks. Let $g(k, n)$ be the number of tilings of a $k \times n$ block with blocks. Nagy asked how one might calculate or estimate $g(k, n)$. He also asked about $d(k, n)$, the number of tilings of a $k \times n$ block which can be produced by a divide-and-conquer method of tiling. Divide-and-conquer tilings are produced by a finite number of applications of the following fundamental operation: Select a sub-block of the tiling and cut it into two smaller blocks. The two tilings of a 3×3 block shown in Figure 1 cannot be formed by the divide-and-conquer method. In addition to these two tilings, there are 320 others, and all of these can be produced by the divide-and-conquer method. Thus, $g(3, 3) = 322$, while $d(3, 3) = 320$. Besides proving these numbers are correct, we will show that both of the sequences $g_3 = (g(3, n): n = 1, 2, \dots)$, $d_3 = (d(3, n): n = 1, 2, \dots)$ satisfy difference equations. Using detailed information about this, it follows that $d(3, n)/g(3, n)$ goes to zero exponentially. Thus, the fact that $d(3, 3)/g(3, 3)$ is nearly 1 might be misleading. We wish to thank the referee who points out the similarity of our problem to certain problems in statistical mechanics [1], [5], [6]. This analogy extends also to our method, which in statistical mechanics is known as the *transfer matrix method*.

What can be said about the numbers $d(k, n)$ and $g(k, n)$? Of course, it follows from the definitions that $g(k, n) = g(n, k)$ and $d(k, n) = d(n, k)$. Also, the following bounds are fairly apparent:

$$2^{nk-k} \leq d(k, n) \leq g(k, n) \leq 2^{2nk-n-k}. \quad (1)$$

To see the left-most inequality, first cut the $k \times n$ block into k horizontal $1 \times n$ blocks. Each of these $1 \times n$ blocks can be cut in 2^{n-1} different ways. Combining these subtilings

FIGURE 1. Two 3×3 non-divide-and-conquer tilings.

gives rise to $(2^{n-1})^k$ different divide-and-conquer tilings of the $k \times n$ block. Since divide-and-conquer tilings are a special case of general tilings, $d(k, n) \leq g(k, n)$. Finally, to see the right-most inequality notice that all possible cuts in the $k \times n$ block consist of $2nk - n - k$ unit line segments. A tiling of the $k \times n$ block corresponds to a unique subset of these $2nk - n - k$ unit segments. Since the total number of subsets of unit segments is $2^{2nk-n-k}$, the inequality follows. Taking (nk) th roots through (1) we get

$$2 \cdot 2^{-1/n} \leq (d(k, n))^{1/kn} \leq (g(k, n))^{1/kn} \leq 4. \quad (2)$$

This leads one to think that the limits

$$\lim_{k, n \rightarrow \infty} (d(k, n))^{1/kn} = \delta, \quad \lim_{k, n \rightarrow \infty} (g(k, n))^{1/kn} = \gamma \quad (3)$$

exist. In fact, they do. This will be shown as a consequence of the following lemma.

LEMMA 1. Suppose $(f(m, n): m, n = 1, 2, \dots)$ is a double sequence of positive numbers such that $\{(f(m, n))^{1/mn}: m, n = 1, 2, \dots\}$ is a bounded set, $f(m, n) \leq f(m+1, n)$, $f(m, n+1)$ for $m, n = 1, 2, \dots$ and

$$(f(m, n))^{uv} \leq f(mu, nv), \quad m, n, u, v = 1, 2, \dots \quad (4)$$

Then,

$$\lim_{m, n \rightarrow \infty} (f(m, n))^{1/mn} = \varrho \quad (5)$$

exists, and $(f(m, n))^{1/mn} \leq \varrho$ for $m, n = 1, 2, \dots$

PROOF. [We are indebted to Harvey Abbott at the University of Alberta in Edmonton for suggesting the following proof based on a similar result due to Fekete.] Since $S = \{(f(m, n))^{1/mn}: m, n = 1, 2, \dots\}$ is bounded, it has a largest limit point ϱ . For each $\varepsilon > 0$ there exist $m_\varepsilon, n_\varepsilon$ such that $\varrho - \varepsilon \leq (f(m_\varepsilon, n_\varepsilon))^{1/m_\varepsilon n_\varepsilon}$. Let $m > m_\varepsilon, n > n_\varepsilon$ and put $m = m_\varepsilon u + r, n = n_\varepsilon v + s$, with $0 \leq r \leq m_\varepsilon, 0 \leq s \leq n_\varepsilon$. Then,

$$(\varrho - \varepsilon)^{m_\varepsilon n_\varepsilon uv} \leq (f(m_\varepsilon, n_\varepsilon))^{uv} \leq f(mu_\varepsilon, vn_\varepsilon) \leq f(mu_\varepsilon + r, vn_\varepsilon + s) = f(m, n).$$

So,

$$(\varrho - \varepsilon)^{1-E} \leq (f(m, n))^{1/mn},$$

where

$$\begin{aligned} E &= (m_\varepsilon us + n_\varepsilon vr + rs)/(m_\varepsilon n_\varepsilon uv + m_\varepsilon us + n_\varepsilon vr + rs) \\ &\leq (m_\varepsilon n_\varepsilon u + m_\varepsilon n_\varepsilon v + m_\varepsilon n_\varepsilon)/m_\varepsilon n_\varepsilon uv \\ &= (u + v + 1)/uv. \end{aligned}$$

Now, as $m = m_\varepsilon u + r, n = n_\varepsilon v + s$ tend to infinity, u, v tend to infinity. Hence, as m, n tend to infinity, E tends to 0 so (5) follows. To see that $(f(m, n))^{1/mn} \leq \varrho$, suppose $\varrho + \varepsilon = (f(m, n))^{1/mn}$ for some m, n and positive ε . Then we have $(\varrho + \varepsilon)^{mn uv} \leq f(mu, nv)$ for $u, v = 1, 2, \dots$; Thus, the subset $\{(f(mu, nv))^{1/mnuv}: u, v = 1, 2, \dots\}$ of S has a limit point $\varrho + \varepsilon$ or greater, a contradiction. This completes the proof.

Since we also plan to use Fekete's lemma, it is quoted here for the reader's convenience. For this and similar results, see Polya and Szego [8], page 171.

LEMMA 2. [Fekete]. *Let $(f(n): n = 1, 2, \dots)$ be an increasing sequence of positive numbers such that $\{(f(n))^{1/n}: n = 1, 2, \dots\}$ is bounded, and $f(m)f(n) \leq f(m+n)$ for $m, n = 1, 2, \dots$. Then, $\lim_{n \rightarrow \infty} (f(n))^{1/n} = \rho$ exists, and $(f(n))^{1/n} \leq \rho$ for large enough n .*

Now, we apply these lemmas to the sequences d and g . Note that if an $m \times n$ block can be tiled with a set M of sub-blocks, then

$$\prod_{u \times v \in M} g(u, v) \leq g(m, n). \quad (6)$$

Similarly, if an $m \times n$ block can be divide-and-conquer tiled with a set M of sub-blocks, then g can be replaced with d in (6). Thus, by cutting a $um \times vn$ block into a rectangular array of uv $m \times n$ sub-blocks and using this as M in (6), we see that both d and g satisfy (4). Also, using (2), it is clear that d and g both satisfy the hypothesis of Lemma 1, so the limits δ and γ defined in (3) exist. Now let $k = 1, 2, \dots$ be fixed, and consider $d_k = (d(k, n): n = 1, 2, \dots)$ and $g_k = (g(k, n): n = 1, 2, \dots)$. Both of these sequences are increasing, and it follows from (2) that

$$(2 - \varepsilon)^k \leq (d(k, n))^{1/n} \leq (g(k, n))^{1/n} < 4^k, \quad n = 1, 2, \dots \quad (7)$$

Also, since a $k \times (m+n)$ block can be cut into a $k \times m$ block and a $k \times n$ block, we have $d(k, m)d(k, n) \leq d(k, m+n)$ and $g(k, m)g(k, n) \leq g(k, m+n)$. Hence, both d_k and g_k satisfy the hypothesis of Lemma 2, so the following limits exist:

$$\lim_{n \rightarrow \infty} (d(k, n))^{1/n} = \delta_k, \quad \lim_{n \rightarrow \infty} (g(k, n))^{1/n} = \gamma_k. \quad (8)$$

Furthermore,

$$(d(k, n))^{1/n} \leq \delta_k, \quad (g(k, n))^{1/n} \leq \gamma_k, \quad n = 1, 2, \dots \quad (9)$$

We will show in subsequent sections that both d_k and g_k are rational sequences, that is, they satisfy linear homogeneous difference equations with constant coefficients. There exist constants a_k, b_k for each k such that $a_k \delta_k^n < d(k, n)$, $b_k \gamma_k^n < g(k, n)$, for $n = 1, 2, \dots$, so,

$$\begin{aligned} (a_k^{1/k})^{1/n} \delta_k^{1/k} &< (d(k, n))^{1/kn} \leq \delta, \\ (b_k^{1/k})^{1/n} \gamma_k^{1/k} &< (g(k, n))^{1/kn} \leq \gamma. \end{aligned} \quad (10)$$

Taking limits as n tends to infinity gives

$$\delta_k^{1/k} \leq \delta, \quad \gamma_k^{1/k} \leq \gamma, \quad k = 1, 2, \dots \quad (11)$$

We suspect, but have not been able to prove, that the sequences $(\delta_k^{1/k}: k = 1, 2, \dots)$ and $(\gamma_k^{1/k}: k = 1, 2, \dots)$ are increasing, and they converge to δ and γ , respectively. Our educated guess about the values of δ and γ is based on the foregoing hypothesis. What have actually been established are lower bounds for δ and γ , and this is based on (11). The lower bounds for δ and γ are obtained by studying the sequences d_k and g_k , and this is done in the next sections.

2. GENERAL TILING OF BLOCKS WITH FIXED WIDTH

Let $k = 1, 2, \dots$ be fixed. Our objective in this section is to show that the sequence $g_k = (g(k, n): n = 1, 2, \dots)$ is rational; that is, there exists a smallest number $\omega_k = \omega$ (called the *order* of g_k) and integers c_i ($i = 1, \dots, \omega$) such that

$$\sum_{i=1}^{\omega} c_i g(k, n + \omega + 1 - i) = g(k, n + \omega + 1), \quad n = 1, 2, \dots \quad (12)$$

Furthermore, we will show how to calculate ω and the coefficients c_i ($i = 1, \dots, \omega$). Saying g_k satisfies (12) is equivalent to saying that the generating function is of the form

$$G_k(z) = \sum_{n=1}^{\infty} g(k, n)z^n = P_k(z)/Q_k(z) \quad (13)$$

where,

$$P_k(z) = \sum_{i=1}^{\omega} \{g(k, i) - c_1 g(k, i-1) - c_2 g(k, i-2) - \dots\} z^i, \quad (14)$$

$$Q_k(z) = 1 - \sum_{i=1}^{\omega} c_i z^i, \quad (15)$$

and P_k, Q_k have no common zeroes.

It turns out that all of the zeroes of $z^{\omega} Q_k(1/z)$ are real, and it follows from the classical theory of difference equations that the largest of these zeroes is the number γ_k defined in the previous section. In fact, if the multiplicity of γ_k is 1,

$$g(k, n) \sim C_k \gamma_k^n,$$

where

$$C_k = \lim_{z \rightarrow 1/\gamma_k} (1 - \gamma_k z) P_k(z)/Q_k(z), \quad (16)$$

but in any case, we always have

$$\lim_{n \rightarrow \infty} (g(k, n))^{1/n} = \gamma_k, \quad (17)$$

since $1/\gamma_k$ is the singularity of G_k nearest to 0.

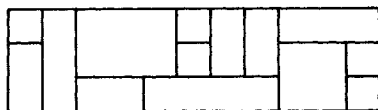


FIGURE 2.

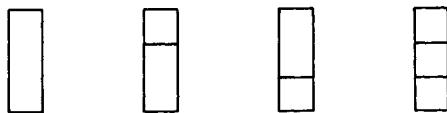


FIGURE 3.

Here is how we find a difference equation satisfied by g_k . To illustrate the method, consider the case $k = 3$. We view a tiling of $3 \times n$ a block such as the one shown in Figure 2 as a sequence of the cross-sections shown in Figure 3. Cross-sections can be pasted together along their common edge in various ways; some of the unit segments on the three unit common edge can be omitted or retained, the rest must always be present. Figure 4 shows all the ways one cross-section can be pasted together with another to its right. This situation can be described by a graph whose vertices are all possible cross-sections, and where there is a directed edge joining vertex i to vertex j for each of the ways of pasting j to the right of i .

Clearly, this relationship is symmetric because every way of joining j to the right of i can be reflected to become a way of joining i on the right of j . The incidence matrix in the case

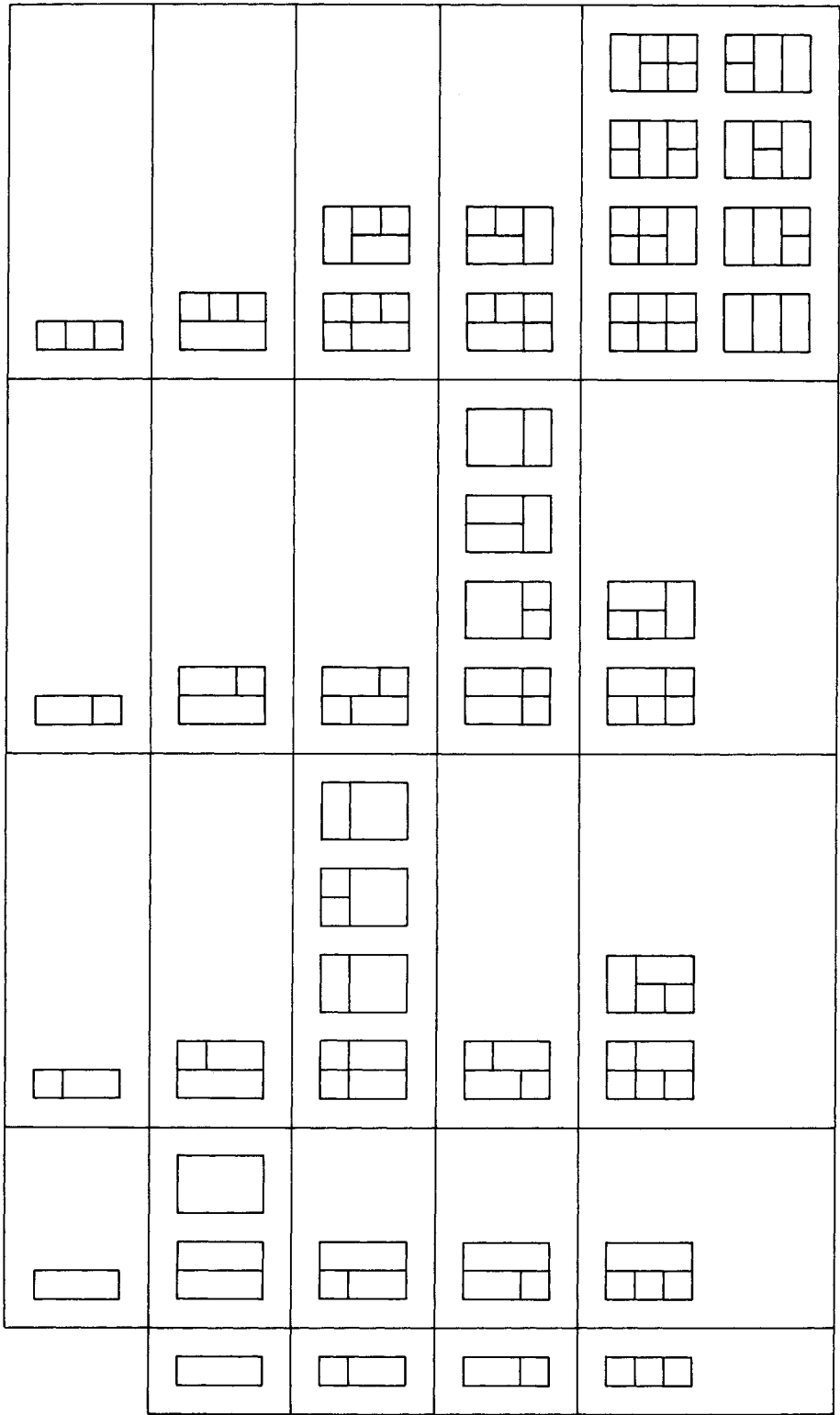


FIGURE 4.

$k = 3$ is

$$\mathbf{M}_3 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 8 \end{bmatrix} \quad (18)$$

By induction on n , it is easy to show that there is a bijection between the collection of paths of length $n - 1$ beginning at i and ending at j and the tilings of a $3 \times n$ block with cross-sections i on the left and j on the right end. Using the well-known result from graph theory which asserts that the number just described is the (i, j) -entry in $\mathbf{M}_3^{n-1}$, we have

$$g(3, n) = \mathbf{V}_1^T \mathbf{M}_3^{n-1} \mathbf{V}_1, \quad (19)$$

where $\mathbf{V}_1 = [1, 1, 1, 1]^T$, and $\mathbf{V}^T$ denotes the transpose of $\mathbf{V}$.

Using (19), it is easy to show that g_3 satisfies the difference equation $g(3, n + 4) = k_3 g(3, n + 3) + k_2 g(3, n + 2) + k_1 g(3, n + 1) + k_0 g(3, n)$, for $n = 1, 2, \dots$, where $\det(\mathbf{M}_3 - z\mathbf{I}) = z^4 - k_3 z^3 - k_2 z^2 - k_1 z - k_0$ is the characteristic polynomial for $\mathbf{M}_3$. Krylov's method [10] sometimes produces a difference equation satisfied by g_k which has order smaller than the degree of the characteristic polynomial. Let $\mathbf{V}_{n+1} = \mathbf{M}_3 \mathbf{V}_n$ for $n = 1, 2, \dots$ with $\mathbf{V}_1$ as above. Then $\mathbf{V}_{n+1} = \mathbf{M}_3^n \mathbf{V}_1$, and $g(3, n) = \mathbf{V}_1^T \mathbf{M}_3^{n-1} \mathbf{V}_1 = \mathbf{V}_1^T \mathbf{V}_n$.

If for some $\omega = 1, 2, \dots$ we can find numbers $c_1, \dots, c_\omega$ such that

$$c_1 \mathbf{V}_1 + \dots + c_\omega \mathbf{V}_\omega = \mathbf{V}_{\omega+1}, \quad (20)$$

then, after multiplying both sides of (20) on the right by $\mathbf{M}_3^n$, we have

$$c_1 \mathbf{V}_{n+1} + \dots + c_\omega \mathbf{V}_{n+\omega} = \mathbf{V}_{n+\omega+1}, \quad n = 1, 2, \dots \quad (21)$$

Hence, g_3 satisfies the difference equation

$$c_1 g(3, n) + \dots + c_\omega g(3, n + \omega) = g(3, n + \omega + 1), \quad n = 1, 2, \dots \quad (22)$$

We now discuss the general case. Suppose that $k = 1, 2, \dots$ is fixed. How does one compute the matrix $\mathbf{M}_k$? First, the cross-sections are represented by binary vectors of length $k + 1$. The vector $(u_0, u_1, \dots, u_k)$ corresponds to the cross-section having a horizontal line segment in position i if and only if $u_i = 1$. This vector is also used to number the cross-sections; a cross-section with vector $(u_0, u_1, \dots, u_k)$ is given the *index*

$$u = \sum_{i=1}^{k-1} u_i 2^i,$$

thus, $\mathbf{M}_k = [m_k(u, v) : 0 \leq u, v < 2^{k-1}]$ is a 2^{k-1} by 2^{k-1} matrix. Note that the bits $u_0 = u_k = 1$ are not used to form the index u . Now suppose that cross-sections with indices u, v are represented by $(u_0, u_1, \dots, u_k), (v_0, v_1, \dots, v_k)$, respectively. In how many ways can u and v be pasted together along a common edge of length k ? Our notation for this is $m_k(u, v)$. To answer this question we need to define a *common block* for u and v . Suppose that there exist i, j such that $0 \leq i < j \leq k$ and $u_i = v_i = u_j = v_j = 1$, while $u_h = v_h = 0$, for $i < h < j$. Then u and v have a common $1 \times (j - i)$ block. Let $e_k(u, v)$ be the number of common blocks of u and v . Then, $m_k(u, v) = 2^{e_k(u, v)}$, because edges shared by common blocks of u and v can be deleted or retained when u and v are pasted together. Of course, edges of blocks which are not common must always be retained. There are $2^{e_k(u, v)}$ different subsets of the $e_k(u, v)$ edges of common blocks of u and v which can be deleted in pasting u to v , and this gives our result.

Let $E_k = [e(u, v) : 0 \leq u, v \leq 2^{k-1}]$, then since $M_k = [2^{e_k(u, v)}]$, computing M_k essentially reduces to computing E_k . But the sequence of matrices $(E_1, E_2, \dots)$ satisfies an unusual recurrence relation; namely,

$$E_{k+1} = \left[\begin{array}{c|c} E_k & E_k - F_k \\ \hline E_k - F_k & J_k + E_k \end{array} \right], \quad (23)$$

where J_k is the all ones matrix having the same dimensions as E_k , and $F_k = [f_k(u, v) : 0 \leq u, v \leq 2^{k-1}]$, with

$$f_k(u, v) = \begin{cases} 1, & \text{if } u = v = 0, \\ 1, & \text{if } 2^{t-1} \leq u, v < 2^t, \text{ for some } t = 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

For example,

$$F_1 = E_1 = [1], \quad M_1 = [2], \quad (25)$$

$$F_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}, \quad (26)$$

$$F_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 8 \end{bmatrix}. \quad (27)$$

We prove (23) in the following equivalent form, and establish that E_k is symmetric.

LEMMA 3. (i) $e_k(u, v) = e_k(u, v)$ for $0 \leq u, v < 2^{k-1}$.

(ii) If $0 \leq u, v < 2^{k-1}$, then $e_{k+1}(u, v) = e_k(u, v)$.

(iii) If $0 \leq u - 2^{k-1}, v - 2^{k-1} < 2^{k-1}$, then $e_{k+1}(u, v) = 1 + e_k(u - 2^{k-1}, v - 2^{k-1})$.

(iv) If $0 \leq u, v - 2^{k-1} < 2^{k-1}$, then $e_{k+1}(u, v) = e_k(u, v - 2^{k-1}) - 1$ when $u = v - 2^{k-1} = 0$, or $2^{t-1} \leq u, v - 2^{k-1} < 2^t$ for some $t = 1, 2, \dots, k-1$; otherwise, $e_{k+1}(u, v) = e_k(u, v - 2^{k-1})$.

PROOF. (i) The definition of $e_k(u, v)$ is symmetric in u and v . (ii) Suppose that $0 \leq u, v < 2^{k-1}$, and $u = (u_0, u_1, \dots, u_k)$, $v = (v_0, v_1, \dots, v_k)$ are the vectors corresponding to u, v , respectively, in the definition of $e_k(u, v)$. Then, the vectors corresponding to u, v in the definition of $e_{k+1}(u, v)$ are $u' = (u_0, u_1, \dots, 0, u_k)$, $v' = (v_0, v_1, \dots, 0, v_k)$, respectively. The common blocks of u', v' are exactly the same as the common blocks of u, v with the possible exception of a common block at the right end, but then this block is only stretched a little longer with the insertion of 0. Hence, $e_{k+1}(u, v) = e_k(u, v)$. (iii) Suppose that $0 \leq u - 2^{k-1}, v - 2^{k-1} < 2^{k-1}$, and $u = (u_0, u_1, \dots, u_k)$, $v = (v_0, v_1, \dots, v_k)$ are the vectors corresponding to $u - 2^{k-1}, v - 2^{k-1}$ respectively in the definition of $e_k(u - 2^{k-1}, v - 2^{k-1})$. Then $u' = (u_0, \dots, 1, u_k)$, $v' = (v_0, \dots, 1, v_k)$ are the vectors corresponding to u, v respectively, in the definition of $e_{k+1}(u, v)$. Clearly u', v' have exactly one more common block than u, v , so $e_{k+1}(u, v) = 1 + e_k(u - 2^{k-1}, v - 2^{k-1})$. (iv) Finally, suppose that $0 \leq u, v - 2^{k-1} < 2^{k-1}$ and let $u = (u_0, u_1, \dots, u_k)$, $v = (v_0, v_1, \dots, v_k)$ be the vectors corresponding to $u, v - 2^{k-1}$ in the definition of $e_k(u, v - 2^{k-1})$. Then the vectors corresponding to u, v in the definition of $e_{k+1}(u, v)$ are $u' = (u_0, \dots, 0, u_k)$, $v' = (v_0, \dots, 1, v_k)$, respectively. If u, v had no common block at their right ends, this

TABLE I
 $g(k, n)$

k/n	1	2	3	4	5	6
1	1	2	4	8	16	32
2	2	8	34	148	650	2,864
3	4	34	322	3,164	31,484	314,662
4	8	148	3,164	70,878	1,613,060	36,911,922
5	16	650	31,484	1,613,060	84,231,996	4,427,635,270
6	32	2,864	314,662	36,911,922	4,427,635,270	535,236,230,270
7	64	12,634	3,149,674	846,280,548	233,276,449,488	64,878,517,290,010
8	128	55,756	31,544,384	19,415,751,782	12,300,505,521,832	
9	256	246,098	315,981,452	445,550,465,628	648,782,777,031,100	
10	512	1,086,296	3,165,414,034	10,225,294,476,962		
11	1,024	4,795,090	31,710,994,234	234,675,373,081,668		
12	2,048	21,166,468	317,682,195,692			
13	4,096	93,433,178	3,182,564,368,244			
14	8,192	412,433,792	31,883,205,466,534			
15	16,384	1,820,570,506	319,408,833,724,882			
16	32,768	8,036,386,492				
17	65,536	35,474,325,410				
18	131,072	156,591,247,016				
19	262,144	691,227,204,226				
20	524,288	3,051,224,496,244				
21	1,048,576	13,468,756,547,882				
22	2,097,152	59,453,967,813,584				
23	4,194,304	262,442,511,046,330				
24	8,388,608					

remains true for u', v' ; in fact, u, v, u', v' have exactly the same set of common blocks in this case. If u, v have a common block at their right ends, this is spoiled in u', v' . This happens when $u = v - 2^{k-1} = 0$, and when $2^{t-1} \leq u, v - 2^{k-1} < 2^t$ for some $t = 1, \dots, k-1$ because in this case $u_t = v_t = 1$, and $u_i = v_i = 0$ for $t < i < k$. This is just the case when a common block at the right ends of u, v is spoiled in u, v . Thus, (iv) is proved, and the entire proof is complete.

Now that we know how to compute $E_k = [e_k(u, v)]$, the computation of $M_k = [2^{e_k(u, v)}]$ is trivial; also, $g(k, n)$ is the sum of all the entries in M_k^{n-1} for $k, n = 1, 2, \dots$. This is the way the entries in Table 1 were computed.

Now we describe the computation of the number γ_k . Since $M = M_k$ is a real symmetric matrix, it has real eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_m$, $m = 2^{k-1}$, and an orthonormal set of real eigenvectors $u_1, \dots, u_m$ corresponding to $\lambda_1, \dots, \lambda_m$. Let $x_1 = V_1^T$ denote the all 1's row vector of dimension m , and $\lambda = \lambda_j$ be an eigenvalue of largest magnitude for which x_1 is not orthogonal to the eigenspace $E(\lambda)$ of λ . Computation shows that if λ is as above, $-\lambda$ is not an eigenvalue of M . We compute λ by the standard power method [2] by generating a sequence $x_1, x_2, \dots$ of vectors so that $x_n = (a_{n,1}, \dots, a_{n,m}) = \xi_n / \|\xi_n\|$, where $\xi_n = x_{n-1}M$. Since x_n converges to an eigenvector corresponding to λ , we have that $\lim_{n \rightarrow \infty} (x_n, x_{n-1}) = 1$, and for each $i = 1, \dots, 2^{k-1}$, $\lim_{n \rightarrow \infty} (a_{n,i} / a_{n-1,i}) = \lambda$. Now recall that $g(k, n) = x_1 M^{n-1} x_1^T = \sum_{i=1}^m a_{n,i}$. We have $a_{n,i} = \lambda a_{n-1,i} (1 + \varepsilon_{n,i})$ where, $\lim_{n \rightarrow \infty} \varepsilon_{n,i} = 0$ for each $i = 1, \dots, 2^{k-1}$. Consequently,

$$\lim_{n \rightarrow \infty} g(k, n) / g(k, n-1) = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^m \lambda (1 + \varepsilon_{n,i}) a_{n-1,i} \right] / \sum_{i=1}^m a_{n-1,i} = \lambda.$$

Hence, $g(k, n)$ is asymptotic to $c\lambda^n$ for some $c > 0$, and we conclude that $\lim_{n \rightarrow \infty} (g(k, n))^{1/kn} = \lim_{n \rightarrow \infty} c^{1/nk} \lambda^{1/k} = \lambda^{1/k}$, and $\lambda = \gamma_k$. The values of γ_k we computed are given in Table 2.

TABLE 2

k	γ_k	δ_k
1	2	2
2	4.41421356237	4.41421356237
3	10.01809243697	9.92806953491
4	22.95084331521	22.41980617414
5	52.75261577530	
6	121.39426448987	
7	279.46925608876	
8	643.47979819355	
9	1481.69586661122	

3. DIVIDE-AND-CONQUER TILING OF BLOCKS WITH FIXED WIDTH

Recall that a *divide-and-conquer* tiling is one obtained by applying the following basic step a finite number of times: Select a sub-block of the tiling, and cut it into two smaller blocks. Let $d(k, n)$ be the number of divide-and-conquer tilings of an $k \times n$ block. For example, it was shown in the section before this one that $g(3, 3) = 322$, and it can be shown that exactly two of these general tilings are not divide-and-conquer tilings (the exceptions are shown in Figure 1), so $d(3, 3) = 320$. Also, it is easy to see that $d(1, n) = g(1, n)$, $d(2, n) = g(2, n)$ for $n = 1, 2, \dots$; furthermore, it follows from the definition that $d(k, n) = d(n, k)$ for $k, n = 1, 2, \dots$.

Let $k = 1, 2, \dots$ be fixed, let $d_k = (d(k, n); n = 1, 2, \dots)$, and let $D_k(z) = d(k, 1)z + d(k, 2)z^2 + \dots$. Our main result in this section is to show that D_k is rational,

that is d_k satisfies a linear homogeneous difference equation with constant coefficients. In fact, the sequence of rational functions $(D_1, D_2, \dots)$ satisfies an unusual recurrence relation. In the rest of this section a divide-and-conquer tiling is just called a *tiling*.

Almost anyone attempting to compute $d(k, n)$ would be led to the sieve formula. Tilings of a $k \times n$ block can be classified in overlapping classes as follows. First, there is exactly one tiling which has no proper sub-blocks, that is, the $k \times n$ block is not cut up at all. All other tilings arise from at least one application of the basic step, and one application of the basic step divides the original block into two sub-blocks which must be subsequently tiled (conquered) by the same procedure. There is one *class* for each way of dividing the $k \times n$ block into exactly two sub-blocks. Moreover, we call the class consisting of the single tiling with no proper sub-blocks, the *special class*. Altogether, the number of non-special classes is $k + n - 2$ and when the special class is added, the total number of classes comes to $k + n - 1$.

Each of the $n + k - 2$ non-special classes for the $k \times n$ block has a *fault line* associated with it in the obvious way. The intersection of several classes $C_1, \dots, C_q$ consists of tilings having all of the lines associated with $C_1, \dots, C_q$ present in the tiling. These lines cut the $k \times n$ block into orderly rows and columns of sub-blocks, and to construct all of the elements of $C_1 \cap \dots \cap C_q$ these sub-blocks must be tiled independently and in all possible ways. Thus, if the lines associated with $C_1, \dots, C_q$ break k, n into compositions $(k_0, \dots, k_i), (n_0, \dots, n_j)$, respectively, where $i + j = q$. Then,

$$|C_1 \cap \dots \cap C_q| = \prod_{r,s} d(k_r, n_s). \quad (28)$$

We now use the sieve formula to get the total number of tilings in terms of intersections of various classes:

$$\begin{aligned} d(k, n) &= 1 + |C_1 \cup \dots \cup C_{k+n-2}| \\ &= 1 - \sum_{q=1}^{k+n-2} (-1)^{q+1} \sum_{1 \leq i_1 < \dots < i_q \leq k+n-2} |C_{i_1} \cap \dots \cap C_{i_q}|. \end{aligned} \quad (29)$$

Taking into account the fact that the intersections $C_{i_1} \cap \dots \cap C_{i_q}$ give rise to all compositions $\mathbf{k} = (k_0, \dots, k_i), \mathbf{n} = (n_0, \dots, n_j)$ of k, n , respectively, with $0 \leq i, j, i + j = q$, we can write (29) as

$$d(k, n) = 1 + \sum_{q=1}^{k+n-2} (-1)^{q+1} \sum_{\mathbf{k}, \mathbf{n}} \prod_{r,s} d(k_r, n_s). \quad (30)$$

See Figure 5 for an example.

$$\begin{aligned} & \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \\ d(2, 3) &= 1 + d(1, 3) + d(2, 1)d(2, 2) + d(2, 2)d(2, 1) \\ & - d(1, 1)d(1, 2)d(1, 1)d(1, 2) - d(1, 2)d(1, 1)d(1, 2)d(1, 1) - d(2, 1)d(2, 1)d(2, 1) \\ & + d(1, 1)d(1, 1)d(1, 1)d(1, 1)d(1, 1)d(1, 1) = 1 + 16 + 16 - 4 - 4 - 8 + 1 = 34 \end{aligned}$$

FIGURE 5. Total number of tilings for $d(2, 3)$.

To show that d_k is rational (for a fixed k) we use (30) to get an equivalent statement involving generating functions. Let $\mathbf{k} = (k_0, \dots, k_i)$ be a composition of k into $i + 1$ positive parts, let $d_k(n) = d(k_0, n) \dots d(k_i, n)$, and define

$$D_{\mathbf{k}}(z) = (D_{k_0} \odot \dots \odot D_{k_i})(z) = \sum_{n=1}^{\infty} d_{\mathbf{k}}(n) z^n. \quad (31)$$

This is the term-by-term product of the generating functions $D_{k_0}, \dots, D_{k_i}$ which is known as the *Hadamard product*. In particular, it is known that the Hadamard product of rational functions is a rational function. (See Klarner [3], or Titchmarsh [9]). Note that

$$(D_{\mathbf{k}}(z))^{j+1} = \sum_{n=j+1}^{\infty} \sum_{\mathbf{n}} d_{\mathbf{k}}(n_0) \dots d_{\mathbf{k}}(n_j) z^n, \quad (32)$$

where the second sum ranges over all compositions $\mathbf{n} = (n_0, \dots, n_j)$ of n into exactly $j + 1$ positive parts. Now multiplying through (30) with z^n , summing over $n = 1, 2, \dots$ and using (32) we get

$$D_{\mathbf{k}}(z) = R_{\mathbf{k}}(z) + (D_{\mathbf{k}}(z))^2 / (1 + D_{\mathbf{k}}(z)), \quad (33)$$

where

$$R_{\mathbf{k}}(z) = \frac{z}{1-z} + \sum_{q=1}^{k-1} (-1)^{q+1} \sum_{\mathbf{k}} D_{\mathbf{k}}(z) / (1 + D_{\mathbf{k}}(z)), \quad (34)$$

where the index of summation in the second sum extends over all compositions $\mathbf{k} = (k_0, \dots, k_q)$ of k into exactly $q + 1$ positive parts for $q = 1, 2, \dots$. Solving for $D_{\mathbf{k}}$ in terms of $R_{\mathbf{k}}$ in (33) yields

$$D_{\mathbf{k}}(z) = R_{\mathbf{k}}(z) / (1 - R_{\mathbf{k}}(z)). \quad (35)$$

We now prove that $D_{\mathbf{k}}$ is rational by induction on k . First, since $d(1, n) = 2^{n-1}$, we have $D_1(z) = z/(1 - 2z)$, so D_1 is rational. Now suppose that $k > 1$, and that $D_1, \dots, D_{k-1}$ are rational. Then $D_{\mathbf{k}}$ is rational for every composition $\mathbf{k}$ of k into more than one parts because $D_{\mathbf{k}}$ is the Hadamard product of rational functions. This means that $R_{\mathbf{k}}$ is a rational expression of rational functions, so $R_{\mathbf{k}}$ is rational. Finally, (35) implies that $D_{\mathbf{k}}$ is rational, so D_i is rational for $i = 1, 2, \dots$.

Now we use (34) and (35) to calculate $D_{\mathbf{k}}$ for a few small values of k . The work involved soon becomes tedious. We have $D_1(z) = z/(1 - 2z)$, so $D_{1,1}(z) = \sum 4^{n-1} z^n = z/(1 - 4z)$. Hence, $D_{1,1}/(1 + D_{1,1}) = z/(1 - 3z)$, thus

$$R_2(z) = \frac{z}{1-z} + \frac{D_{1,1}(z)}{1 + D_{1,1}(z)} = \frac{2z - 4z^2}{1 - 4z + 3z^2}, \quad (36)$$

$$D_2(z) = \frac{R_2}{1 - R_2} = \frac{2z - 4z^2}{1 - 6z + 7z^2}. \quad (37)$$

This agrees with the observation that $D_2 = G_2$ for purely combinatorial reasons. To calculate D_3 we need R_3 , and this requires $D_{1,1,1}$ and $D_{1,2} = D_{2,1}$. But $D_{1,1,1} = \sum 8^{n-1} z^n = z/(1 - 8z)$, so $D_{1,1,1}/(1 + D_{1,1,1}) = z/(1 - 7z)$. Also, $D_{1,2} = \sum 2^{n-1} d(2, n) z^n = \frac{1}{2} D_2(2z) = (2z - 8z^2)/(1 - 12z + 28z^2)$, so $D_{1,2}/(1 + D_{1,2}) = (2z - 8z^2)/(1 - 10z + 20z^2)$. Using these functions in (34) gives

$$\begin{aligned} R_3(z) &= \frac{z}{1-z} + \frac{D_{1,2}(z)}{1 + D_{1,2}(z)} + \frac{D_{2,1}(z)}{(1 + D_{2,1}(z))} - \frac{D_{1,1}(z)}{(1 + D_{1,1,1}(z))} \\ &= \frac{z}{1-z} + \frac{2(2z - 8z^2)}{1 - 10z + 20z^2} - \frac{z}{1 - 7z} \\ &= \frac{4z - 54z^2 + 216z^3 - 232z^4}{1 - 18z + 107z^2 - 230z^3 + 140z^4}. \end{aligned} \quad (38)$$

Putting this into (35) with $k = 3$ gives

$$D_3(z) = \frac{4z - 54z^2 + 216z^3 - 232z^4}{1 - 22z + 161z^2 - 446z^3 + 372z^4} = \sum_{n=1}^{\infty} d(3, n)z^n. \quad (39)$$

Cross-multiplying in the second equality in (39) and equating the coefficients of z^n for $n = 1, 2, \dots$ yields the values of $d(3, n)$ given in Table 3.

Finally, we calculate D_4 . This requires R_4 , so putting $k = 4$ in (34) we get:

$$R_4 = \frac{z}{1-z} + \frac{D_{1,1,1,1}}{1+D_{1,1,1,1}} + \frac{2D_{1,3}}{1+D_{1,3}} + \frac{D_{2,2}}{1+D_{2,2}} - \frac{3D_{1,1,2}}{1+D_{1,1,2}}. \quad (40)$$

Next,

$$\frac{D_{1,1,1,1}}{1+D_{1,1,1,1}} = \frac{z}{1-15z}, \quad (41)$$

$$\frac{2D_{1,3}}{1+D_{1,3}} = \frac{8z - 216z^2 + 1728z^3 - 3712z^4}{1 - 40z + 536z^2 - 2704z^3 + 4096z^4}, \quad (42)$$

$$\frac{D_{2,2}}{1+D_{2,2}} = \frac{4z - 52z^2 + 112z^3}{1 - 25z + 151z^2 - 231z^3}, \quad (43)$$

$$\frac{-3D_{1,1,2}}{1+D_{1,1,2}} = \frac{-6z + 48z^2}{1 - 22z + 96z^2}, \quad (44)$$

and substituting these expressions into (40) gives $R_4(z) = S_4(z)/T_4(z)$, where

$$S_4(z) = 8z - 740z^2 + 29570z^3 - 668434z^4 + 9405918z^5 - 85489802z^6 + 504550824z^7 \\ - 1895296288z^8 + 43044515008z^9 - 5298109440z^{10} + 2667442176z^{11}, \quad (45)$$

$$T_4(z) = 1 - 103z + 4620z^2 - 118442z^3 + 1915933z^4 - 20386823z^5 + 144482646z^6 \\ - 676240640z^7 + 2028629248z^8 - 3671450208z^9 + 3555657216z^{10} \\ - 1362493440z^{11}. \quad (46)$$

Now since $D_4 = R_4/(1 - R_4) = S_4/(T_4 - S_4)$, we just have to subtract S_4 from T_4 to get the denominator of D_4 which is

$$T_4 - S_4 = 1 - 111z + 5360z^2 - 148012z^3 + 2584367z^4 - 29792741z^5 + 229972448z^6 \\ - 1180791472z^7 + 3923925536z^8 - 7975865216z^9 + 8853766656z^{10} \\ - 4029935616z^{11}; \quad (47)$$

Thus, $D_4 = S_4/(T_4 - S_4)$ where S_4 is given in (45) and $T_4 - S_4$ in (47). We used this quotient to calculate the fourth column in Table 3. Singularities of D_1 , D_2 , D_3 , and D_4 nearest 0 can be calculated, and one finds that $\gamma_1 = \delta_1 = 2$, $\gamma_2 = \delta_2 = 4.414213562$, $\gamma_3 = 10.018092$, $\delta_3 = 9.928069$, $\gamma_4 = 22.950843$, $\delta_4 = 22.1419806$. Thus, δ_4 differs from γ_4 by a little more than $1/2$. The work to calculate δ_5 was too much for us.

4. CLOSING REMARKS

Generating functions for $t(m, n)$, the number of ways to tile an $m \times n$ rectangle with $mn/2$ dominoes, are also rational for each fixed m (see Klarner and Polack [4]). One could use the method given in the foregoing section to find the generating function for the number of divide-and-conquer domino tilings of $m \times n$ rectangles with fixed width m . One can obtain the number of fault-free tilings using this information. Thus, Ronald Graham's result

TABLE 3
 $d(k, n)$

k/n	1	2	3	4
1	1	2	4	8
2	2	8	34	148
3	4	34	320	3,118
4	8	148	3,118	68,480
5	16	650	30,752	1,525,558
6	32	2,864	304,618	34,151,364
7	64	12,634	3,022,112	765,712,886
8	128	55,756	29,998,462	17,175,137,464
9	256	246,098	297,826,016	423,992,185,518
10	512	1,086,296	2,956,964,026	12,884,979,890,036
11	1,024	4,795,090	29,358,308,384	456,207,036,604,010
12	2,048	21,166,468	291,482,551,534	
13	4,096	93,433,178	2,893,943,161,568	
14	8,192	412,433,792	28,731,873,679,114	
15	16,384	1,820,570,506	285,256,299,193,376	
16	32,768	8,036,386,492		
17	65,536	35,474,325,410		
18	131,072	156,591,247,016		
19	262,144	691,227,204,226		
20	524,288	3,051,224,496,244		
21	1,048,576	13,468,756,547,882		
22	2,097,152	59,453,967,813,584		
23	4,194,304	262,442,511,046,330		
24	8,388,608			

concerning the existence of fault-free domino tilings for all large $m \times n$ rectangles (m, n both large) could be obtained by counting.

The complexity of computing the sequence d_k , discussed in the foregoing section, is probably closely related to the degree $\omega(k)$ of the sequence, the first few degrees being $\omega(1) = 1, \omega(2) = 2, \omega(3) = 4, \omega(4) = 11$, and $\omega(5) = 30$. Suppose that $D_k = S_k/T_k$ with S_k, T_k polynomials having no common zeroes, the degree of T_k being $\omega(k)$. If all the zeroes of $T_1, \dots, T_{n-1}$ are distinct, it follows from (34) and (35) that

$$2\omega(n) - 1 = \sum_{k_1 + 2k_2 + \dots = n} \prod_{i=1}^n \binom{\omega(i) + k_i - 1}{k_i}, \quad (48)$$

where the index of the sum extends over all partitions $(k_1, k_2, \dots)$ of n ; that is, $k_1, k_2, \dots$ are non-negative integers so that $k_1 + 2k_2 + \dots = n$. Ira Gessel at MIT has pointed out that the generating function for ω is very similar to the one for classes of isomorphic homeomorphically reduced trees with n vertices. One ought to be able to work out an asymptotic formula for $\omega(n)$ using known methods. We leave this as an interesting open problem.

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